AP PRECALCULUS

Unit 1 • Worksheet 1.6A

# Worksheet 1.6A • Polynomial Functions and End Behavior

Name: _____ Date: _____

**Directions**

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

## Question 1

NO CALCULATOR

$\lim_{x \to \infty} \left(-4x^3 + x^2\right)$ is

(A) $\infty$

(B) $-\infty$

(C) $-4$

(D) $0$

## Question 2

NO CALCULATOR

Let $p(x) = (2 - x)^3$. What is $\lim_{x \to -\infty} p(x)$?

(A) $-\infty$

(B) $\infty$

(C) $8$

(D) $0$

## Question 3

NO CALCULATOR

Which polynomial increases without bound at both ends?

(A) $x^3 + 1$

(B) $-x^4$

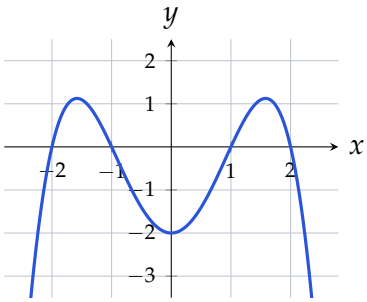
(C) $5x - x^2$

(D) $2x^6 - x$

### Question 4

NO CALCULATOR

The graph of a polynomial $r$ with zeros $\pm 1$, $\pm 2$ and $r(0) = -2$ is shown. Which could be $r(x)$?




(A) $x^4 - 5x^2 + 4$
(B) $-x^3 + 4x$
(C) $-\frac{1}{2}\left(x^4 - 5x^2 + 4\right)$
(D) $\frac{1}{2}\left(x^4 - 5x^2 + 4\right)$

### Question 5 (Free response)

NO CALCULATOR

For each polynomial, write limit statements for the end behavior. (a) $f(x) = -2x^4 + 3x^3 + x - 7$ (b) $g(x) = x^2(3 - x)(x + 1)^2$ (c) $h(x) = (2x - 1)^3(x + 5)^2$